

Feasibility Study

Feasibility study

Feasibility study is a detailed analysis that considers all the critical aspects of a proposed project to determine the likelihood of it succeeding.

Types Of Feasibility Studies



1. Legal
Feasibility



2. Economic
Feasibility



3. Technical
Feasibility



4. Operational
Feasibility



5. Scheduling
Feasibility

Legal Feasibility study

- is one of the most important types of feasibility studies in project management because, even if money is available, each country's laws must allow the legal implementation of the project.
- Several other legal requirements include data protection acts, social media laws, zoning laws, and so on.
- Certain types of jobs are permitted in certain areas.
- As a result, with legal feasibility in place, prior steps can be taken after app development for any legal implications.

Economic feasibility

Economic feasibility: a process of identifying the financial benefits and costs associated with a development project.

Often referred to as **cost-benefit analysis**.



Determining Project Benefits

Tangible benefits refer to items that can be measured in dollars and with certainty.

Examples include:

- reduced personnel expenses,
- lower transaction costs
- higher profit margins.

Determining Project Benefits (Cont.)

Most tangible benefits will fit within the following categories:

- Cost reduction and avoidance
- Error reduction
- Increased flexibility
- Increased speed of activity
- Improvement of management planning and control
- Opening new markets and increasing sales opportunities

Determining Project Benefits (Cont.)

Intangible benefits are benefits derived from the creation of an information system that cannot be easily measured in dollars or with certainty.

May have direct organizational benefits, such as

- the improvement of employee morale.

May have broader societal implications, such as

- the reduction of waste creation or resource consumption.



Determining Project Costs

Tangible costs: a cost associated with an information system that can be measured in dollars and with certainty.

The development tangible costs include:

- Hardware costs,
- Labor costs, or
- Operational costs including employee training and building renovations.



Determining Project Costs(cont.)

Intangible costs: a cost associated with an information system that cannot be easily measured in terms of dollars or with certainty.

Intangible costs can include:

- Loss of customer goodwill,
- Employee morale,
- Operational inefficiency.



Determining Project Costs(cont.)

One-time cost: a cost associated with project start-up and development or system start-up.

These costs encompass activities such as:

- Systems development,
- New hardware and software purchases,
- User training,
- Site preparation, and
- Data or system conversion.



Determining Project Costs(cont.)

Recurring cost: a cost resulting from the ongoing evolution and use of a system.

Examples of these costs include:

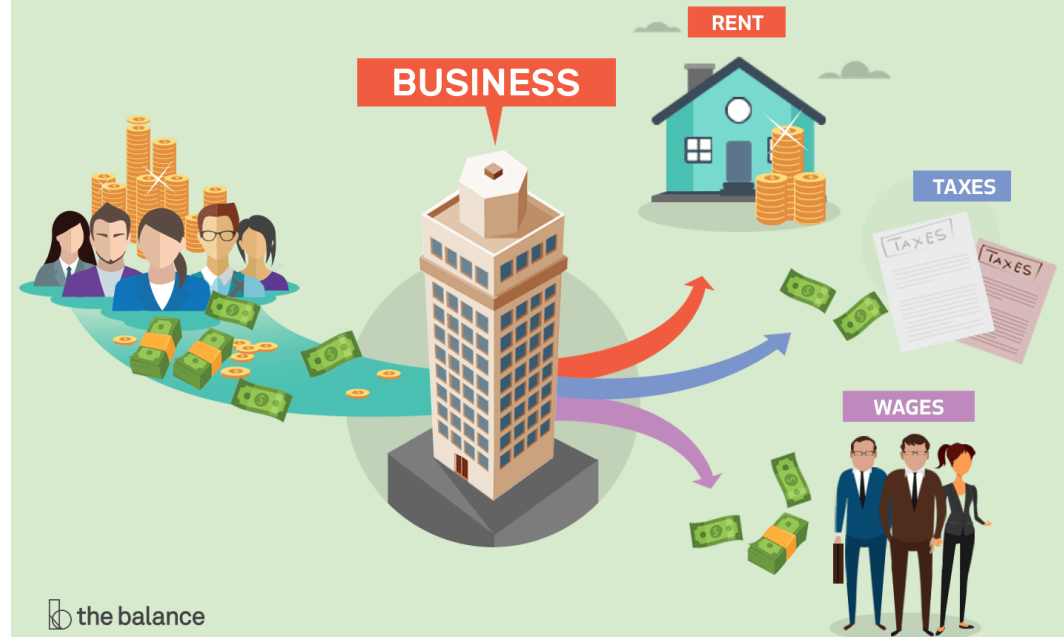
- Application software maintenance,
- Incremental data storage expenses,
- Incremental communications,
- New software and hardware leases, and
- Supplies and other expenses (i.e. paper, forms, data center personnel).

Cash Flow Analysis



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Cash Flow - How It Works to Keep Your Business Afloat



the balance



Cash Flow Analysis and Measures

A cash flow analysis determines a project working budget .

Cash flow analysis helps you understand if your business can pay its bills and generate enough cash to continue operating indefinitely.

Long-term negative cash flow situations can indicate a potential bankruptcy while continual positive cash flow is often a sign of good things to come.

Types of Cash Flow



Operating cash flow



Investing cash flow



Financing cash flow

Cash Flows From Operations (CFO)

- Cash flow from operations (CFO), or operating cash flow, describes money flows involved directly with the production and sale of goods from ordinary operations. CFO indicates whether or not a company has enough funds coming in to pay its bills or operating expenses.

Operating cash flow = cash received from sales - operating expenses.

- Operating cash flow is recorded on a company's cash flow statement, indicates whether a company can generate enough cash flow to maintain and expand operations, and shows when a company may need external financing for capital expansion.

Cash Flows From Investing (CFI)

- Cash flow from investing (CFI) or **investing cash** flow reports how much cash has been generated or spent from various investment-related activities in a specific period.
- Investing activities include purchases of speculative assets, investments in securities, or sales of securities or assets.

Cash Flows From Financing (CFF)

- Cash flows from financing (CFF), or **financing cash flow**, shows the net flows of cash used to fund the company and its capital.
- Cash flow from financing activities provides investors insight into a company's financial strength and how well its capital structure is managed.

Evaluating a project's worth

Return on Investment (ROI): it is a calculation that measures the average rate of return earned on the money invested in the project. ROI is a simple calculation that divides the project's net benefits (total benefits - total costs) by the total costs.

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

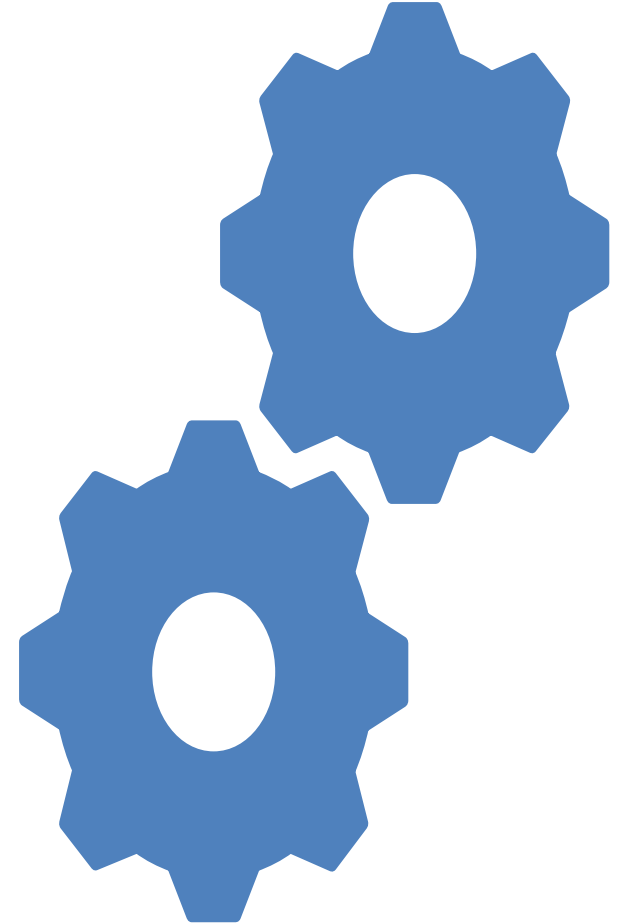
ROI Example

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Costs	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total Benefits – Total Costs)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

$$\text{ROI} = \frac{152,000 - 138,000}{138,000} = \frac{14,000}{138,000} = 10.14\%$$

Technical Feasibility study

- Analyses and evaluates the project's current resources, including hardware and software along with the technical requirements of the proposed system.
- A technical feasibility study gives a report on whether there exist required resources and new technologies which will be used for proposed software development.
- A technical feasibility study examines the technical skills of the software development team, the viability of using current technology, the ease of maintaining and upgrading the technology of choice, and other factors.
- Technical feasibility aspects, particularly in software project development must be checked because they are critical to successful delivery.



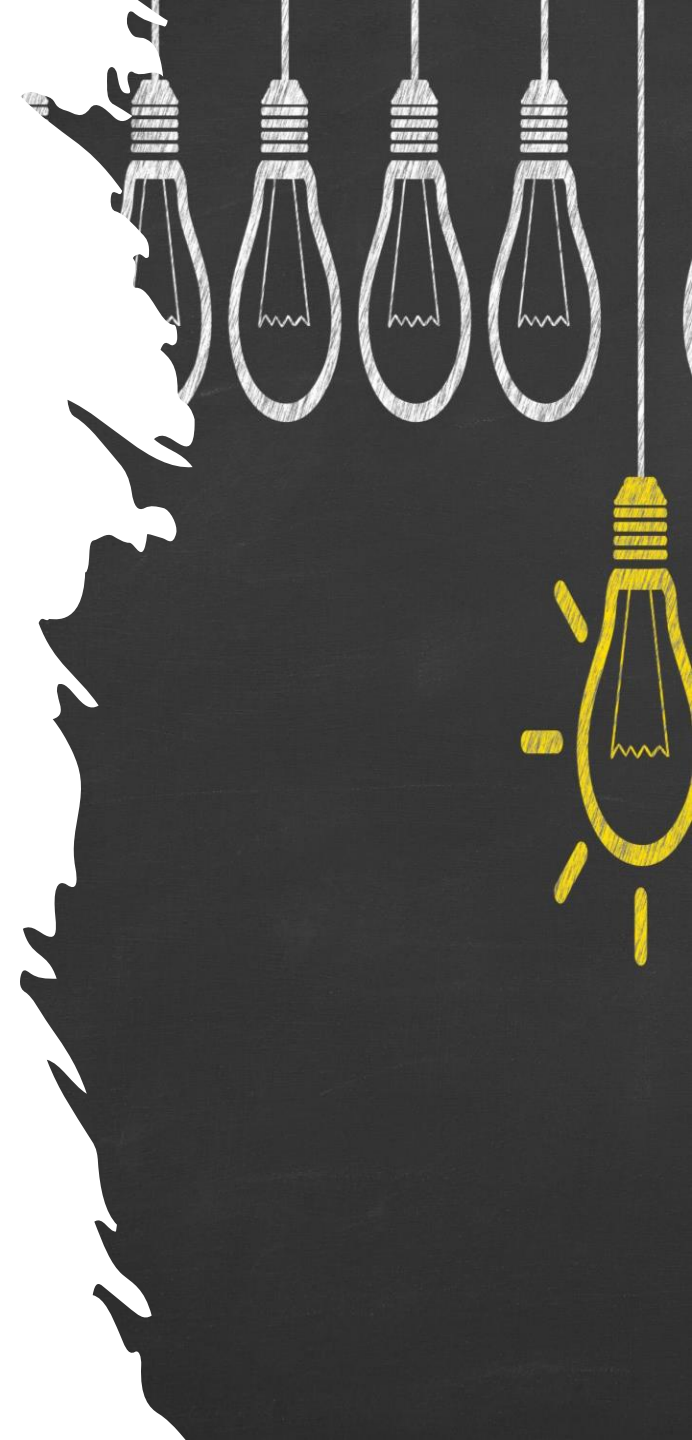
Technical Feasibility Study (Cont.)

Four general rules emerged as technical feasibility assessments:

1. *Larger projects are riskier than smaller projects.*
2. *A system in which the requirements are easily obtained and highly structured will be less risky than one in which requirements are messy, ill structured, ill defined, or subject to the judgment of an individual.*
3. *The development of a system employing commonly used or standard technology will be less risky than one employing novel or nonstandard technology.*
4. *A project is less risky when the user group is familiar with the systems development process and application area than if unfamiliar.*

Operational Feasibility study

- Operational feasibility investigates how a new project will affect your company's daily processes, what procedures should be implemented, and what efforts should be made to keep it running.
- Assume you're launching a global e-commerce platform. Then, in each country, you must have local warehouses, service teams, and, in some cases, even local websites.
- Though the initial idea appears commercially appealing and technically feasible it is extremely difficult to realize operationally and may make no sense at all.





Scheduling Feasibility Study

- This assessment is the most important for project success; after all, a project will fail if not completed on time.
- **In scheduling feasibility, an organization estimates how much time the project will take to complete.**

Feasibility Study Report

The result of the software development team's feasibility analysis is called the feasibility report. It includes recommendations on how software development should be continued in the future by the software development team. This report may also include information about changes to the software scope, technical skills budget, and schedule, as well as other suggestions for system requirements.

